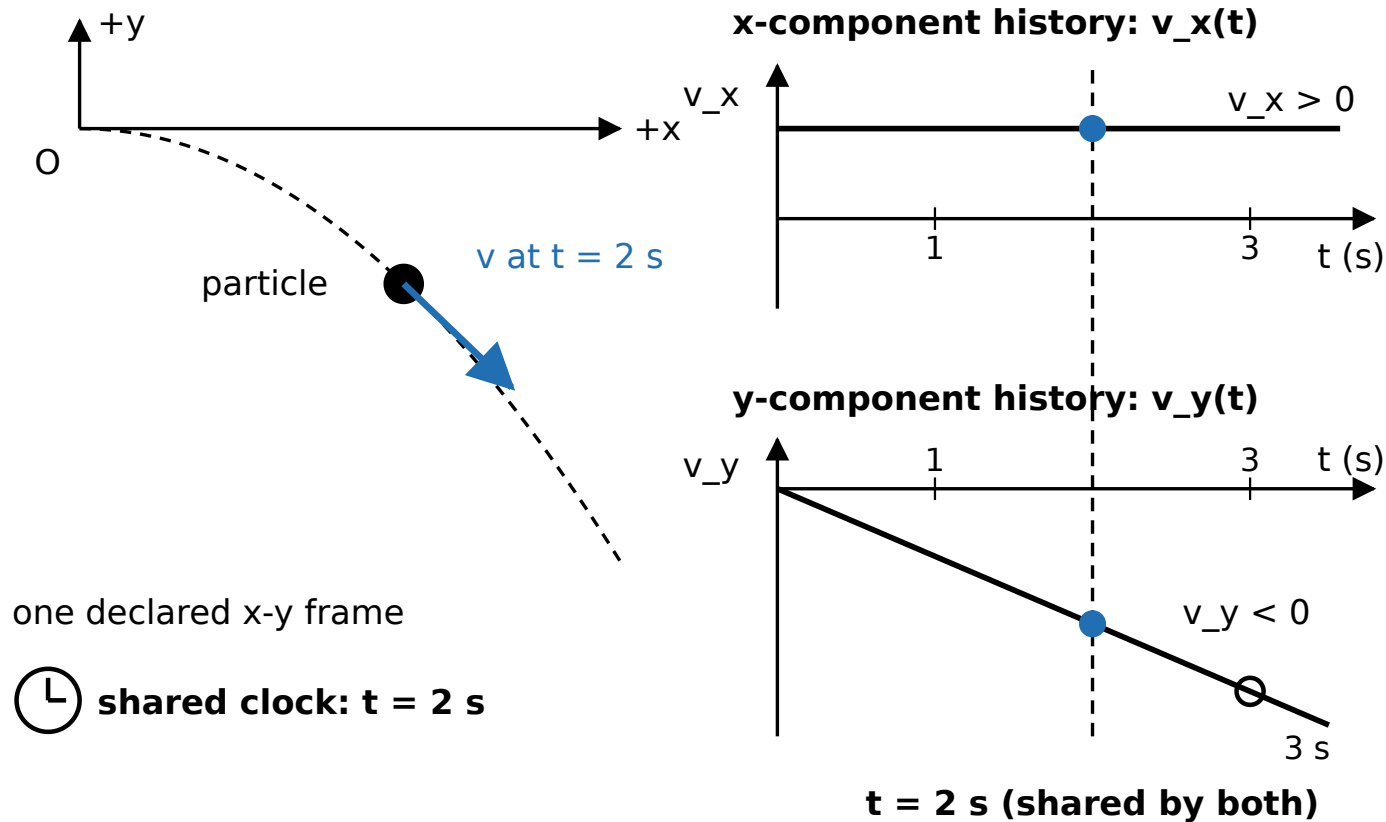


**Owner-approved ExamSIDE question-demand
slice for plane motion, now semantically:
Reconstruction**

Split one plane-motion event into independent x/y components with one clock

Predict

If x and y components are independent, does that mean the object has two different clocks?



✓ ($v_x(2$ s), $v_y(2$ s)): simultaneous, one valid plane state

✗ ($v_x(2$ s), $v_y(3$ s)): non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Attempt

A two-column x/y setup with one shared t.

Reconstruct

1. What must be declared before any component can have a sign?
2. What separates between axes, and what must stay common because both descriptions belong to one event?
3. Which x and y values may be recombined into one state?

Diagnose

- At one instant 2 s after launch, may the x equation use $t=2$ s while the y equation uses another time?
- May $x(2\text{ s})$ be combined with $y(3\text{ s})$ to describe the particle's position?

Repair

- No. Independence separates the axis equations; it does not give the same object two clocks.
- No. A plane state is a simultaneous pair. Recombine $x(t)$ and $y(t)$ only at the same t in the same frame.

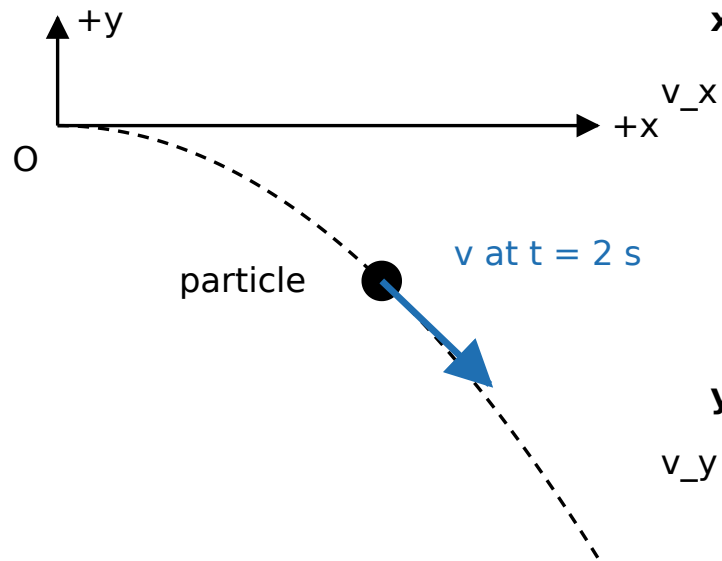
What a complete answer does

Axes are kept separate while time/frame remain shared.

- Separate x and y equations; same t for both.

The step you rebuilt

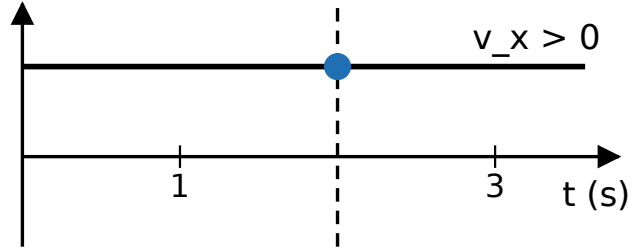
Perpendicular components may be solved separately because each axis equation contains only that axis's quantities, but they still describe one object at one physical time.



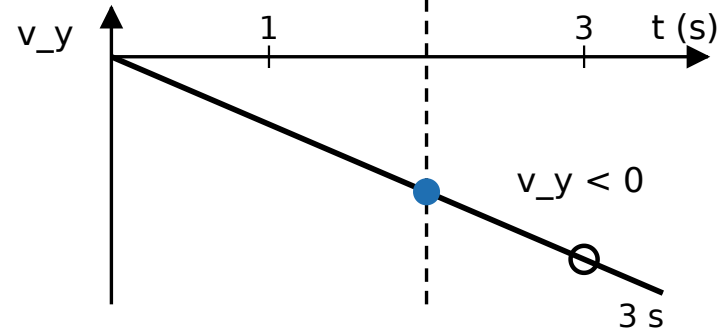
one declared x-y frame

⌚ shared clock: $t = 2 \text{ s}$

x-component history: $v_x(t)$



y-component history: $v_y(t)$



$t = 2 \text{ s}$ (shared by both)

✓ $(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Boundary test

If $a_x=0$ but a_y is nonzero, may v_x stay constant while v_y changes?

Boundary answer

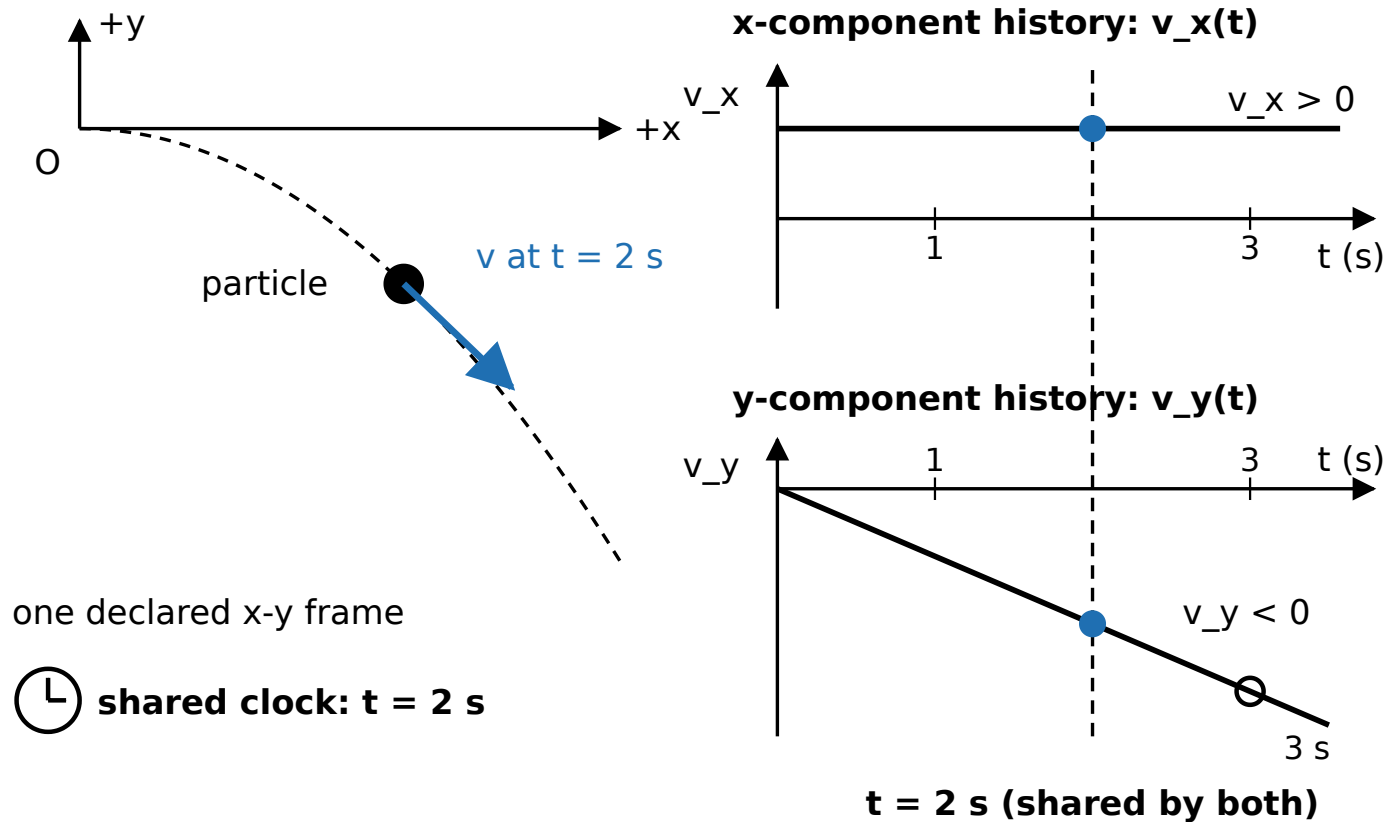
Yes. That is exactly what independent component evolution permits.

One component can change while the other does not.

Apply constant-acceleration kinematics component by component

Predict

If a_x is nonzero and a_y is zero, which velocity component changes with time?



✓ **$(v_x(2\text{ s}), v_y(2\text{ s}))$: simultaneous, one valid plane state**

✗ **$(v_x(2\text{ s}), v_y(3\text{ s}))$: non-simultaneous, rejected (no single instant)**

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Attempt

Two correct component constant-acceleration equations using one t .

Reconstruct

1. Before using a constant-acceleration formula, what must be true about that axis's acceleration component?
2. What does the 1D relation become when written only along x , and what changes for y ?
3. If one component equation finds the event time, what time must the other component use?

Diagnose

- If $\mathbf{a} = (10 \mathbf{i} + 4 \mathbf{j}) \text{ m/s}^2$, should both component equations use $a = \sqrt{116}$?
- If a_x is constant but a_y changes with time, may the same constant-acceleration displacement formula be used exactly on y over the whole interval?
- If x -motion says an event occurs at $t = 3 \text{ s}$, what time belongs in the y equation for that same event?

Repair

- No. Use $a_x = 10$ only in the x equation and $a_y = 4$ only in the y equation; the magnitude is not either component.

- No. Model validity is checked per axis. A constant a_x does not make a_y constant.
- Use $t=3$ s. Both components describe the same event and therefore share one clock.

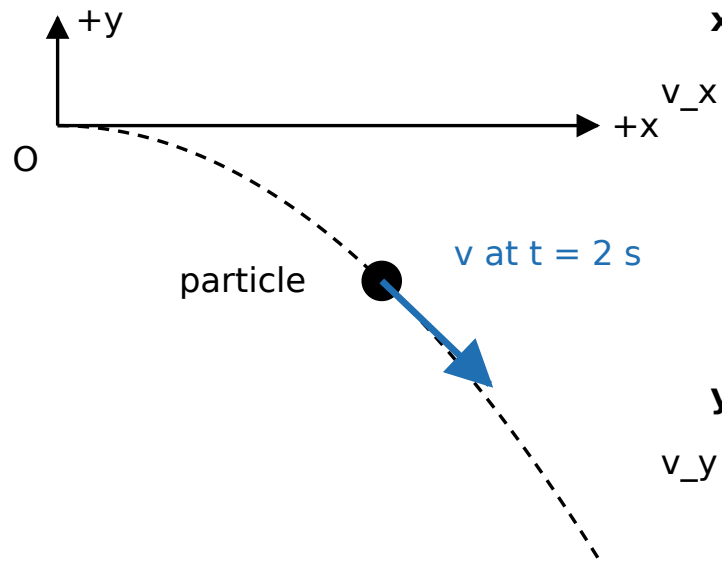
What a complete answer does

Each axis uses only its own signed initial values and acceleration while sharing the same time.

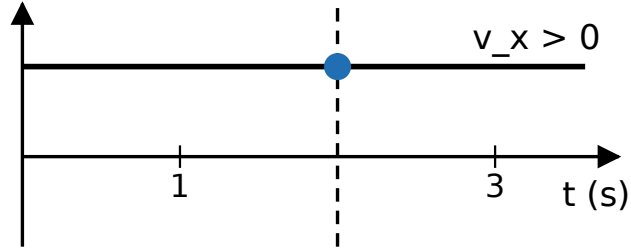
- $v_x = u_x + a_x t$ and $v_y = u_y + a_y t$ with one t .

The step you rebuilt

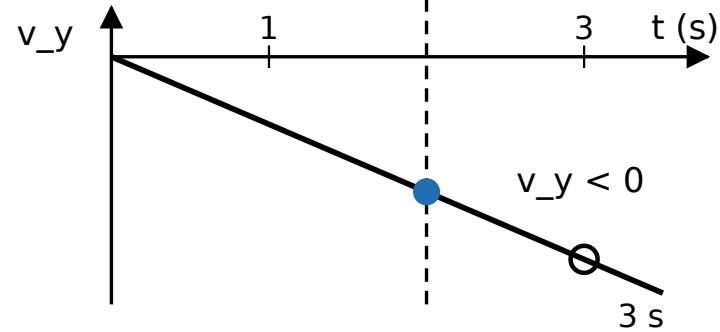
The one-dimensional constant-acceleration equations can be applied to each perpendicular component independently when a_x and a_y are each constant over the same interval.



x-component history: $v_x(t)$



y-component history: $v_y(t)$



one declared x-y frame

⌚ **shared clock: $t = 2$ s**

$t = 2$ s (shared by both)

✓ **$(v_x(2 \text{ s}), v_y(2 \text{ s}))$: simultaneous, one valid plane state**

✗ $(v_x(2 \text{ s}), v_y(3 \text{ s}))$: non-simultaneous, rejected (no single instant)

Keep independently evolving x and y descriptions attached to one physical object, one declared frame and one shared clock, so only simultaneous component values are recombined into a plane state.

Boundary test

If a_y changes with time while a_x stays constant, may the constant-acceleration equations still be exact on both axes for the whole interval?

Boundary answer

No. The x relation may remain exact, but the y constant-acceleration relation is not exact over that interval.

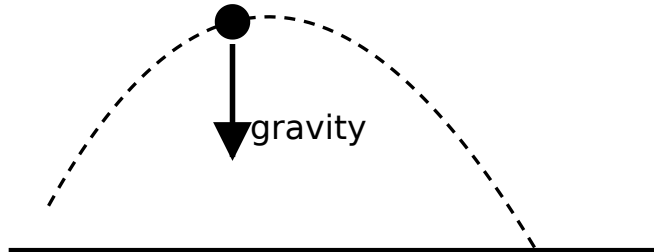
Model validity is checked per component.

Select the gravity-only projectile model before using projectile consequences

Predict

Immediately after a horizontal launch, which acceleration component is zero in the ideal model?

A: launched, then free flight



After release, acting on the object:

- gravity (Earth on object), down
- drag: neglected

Gravity only? yes

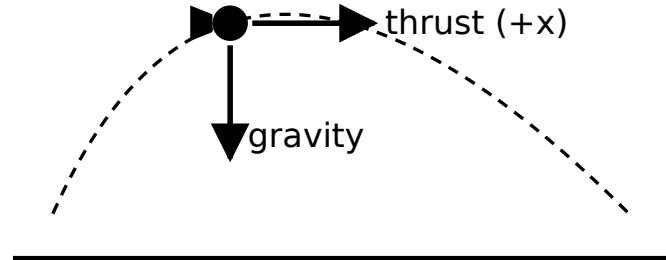
$$a_x = 0$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: VALID

$a_x = 0$ and $a_y = -g$ both hold after release.

B: launched, motor stays on



After release, acting on the object:

- gravity (Earth on object), down
- thrust (motor on object), horizontal
- drag: neglected

Gravity only? no (thrust retained)

$$a_x \neq 0 \text{ (from the thrust)}$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: REJECTED

Active thrust violates $a_x = 0$.
Use the general 2D
constant-acceleration model.

Make the ideal projectile specialization conditional on the interactions that remain after release: gravity-only free flight licenses $a_x=0$ and $a_y=-g$; retained thrust, significant drag or contact requires a more general model.

Attempt

A model declaration and component-equation setup for a horizontal or oblique projectile.

Reconstruct

1. What forces remain after release in the idealized problem?
2. What are the horizontal and vertical component equations, and what clock do they share?
3. What exact component condition identifies the event the question asks about?
4. If the event is the apex, what becomes zero and what definitely does not?
5. Does launch height equal landing height? If not, which shortcut becomes invalid?

Diagnose

- If $a_x=0$ after release, can v_x still be nonzero?
- One second after a horizontal launch, is v_y still zero under gravity?

Repair

- Yes. Zero horizontal acceleration means horizontal velocity stays constant, not that it becomes zero.
- No. $u_y=0$ only at release; $a_y=-g$ makes $v_y=-gt$ afterward.

What a complete answer does

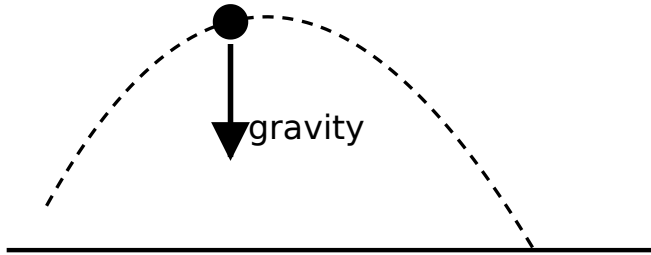
The learner selects gravity-only free flight, uses $a_x=0/a_y=-g$ and one common time before calculating any named projectile quantity.

- State assumptions, set component accelerations, then solve x/y with one t.

The step you rebuilt

After release in the ideal near-Earth model, projectile motion is not a new set of laws: it is the same component kinematics with $a_x=0$ and $a_y=-g$ (for +y upward).

A: launched, then free flight



After release, acting on the object:

- gravity (Earth on object), down
- drag: neglected

Gravity only? yes

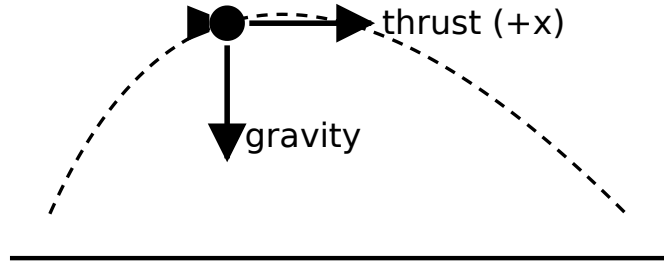
$$a_x = 0$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: **VALID**

$a_x = 0$ and $a_y = -g$ both hold after release.

B: launched, motor stays on



After release, acting on the object:

- gravity (Earth on object), down
- thrust (motor on object), horizontal
- drag: neglected

Gravity only? no (thrust retained)

$$a_x \neq 0 \text{ (from the thrust)}$$

$$a_y = -g \text{ (+y up)}$$

Ideal projectile model: **REJECTED**

Active thrust violates $a_x = 0$.
Use the general 2D
constant-acceleration model.

Make the ideal projectile specialization conditional on the interactions that remain after release: gravity-only free flight licenses $a_x=0$ and $a_y=-g$; retained thrust, significant drag or contact requires a more general model.

Boundary test

A launched object continues under significant horizontal thrust after release. May the standard $a_x=0$ projectile specialization be used?

Boundary answer

No. The object is not in gravity-only free flight; its horizontal acceleration is not zero under that model.

Projectile equations are conditional consequences of a declared force/model state.